

The Deepfake Discount: Believing Erotic Images Are AI-Generated Reduces Affective Responses

Ana Neves¹, Marco Viola², Marco Sperduti³, Guido Corradi⁴, Antonio Olivera-La Rosa⁴, Alessandro Ansani⁴, Marco Marini⁴, and Dominique Makowski^{1,8}

¹School of Psychology, University of Sussex

²Department of Philosophy, Communication, and Performing Arts, Roma Tre University

³Tor Vergata University of Rome

⁴Department of Social Psychology and Organization, Universidad Nacional de Educación a Distancia

⁵Department of Psychology and Social Sciences, Universidad Católica Luis Amigó

⁶Department of Music, Art and Culture Studies, University of Jyväskylä

⁷Institute of Cognitive Sciences and Technologies

⁸Sussex Centre for Consciousness Science, University of Sussex

The proliferation of AI-generated content raises the question of whether beliefs about a stimulus's origin alter emotional responses to it. Across two studies, we conceptually replicated and extended Marini et al. (2024) by presenting heterosexual participants (Study 1: $N = 705$; Study 2: $N = 261$) with erotic (Studies 1 and 2) and non-erotic (Study 1) images from validated databases (NAPS and NAPS ERO). Unbeknownst to participants, all stimuli were real photographs, each randomly labelled as "Photograph" or "AI-generated". Participants rated sexual arousal, enticement, and valence for each image, and later in a second phase judged their agreement with the labels, allowing us to assess manipulation effects. Images labelled as AI-generated received lower ratings, but this effect was small and driven primarily by male participants. Gender differences were robust yet context-dependent: males yielded more robust differences, partially driven by a higher baseline arousal response. However, couple-based stimuli (Study 2), that females rated as more arousing, narrowed the gender gap found in Study 1. Critically, the effect of the labels interacted with participant's own beliefs: photograph-labelled images were rated more arousing than AI-labelled only when participants' subsequent belief ratings were congruent with the label, and the difference between labels reversed when opposite beliefs were later reported, suggesting a complex relationship between emotion and reality beliefs. Individual-level differences in AI attitudes, pornography consumption, and sexual activity showed no reliable modulation of the labels effect. These findings suggest that the anti-AI bias is best understood as a belief-sensitive authenticity effect rather than a uniform response to AI labels, with implications for research using synthetic stimuli and for policies mandating disclosure of AI-generated content.

Keywords: AI-generated, beliefs, gender differences, affective ratings

Introduction

Generative artificial intelligence (genAI, Fisher, 2025) enables the creation of media content (e.g., images, audios, and videos) that are increasingly indistinguishable from authentic ones (Miller et al., 2023; Velásquez-Salamanca et al., 2025). The increasing availability of such content, often called deepfakes, has led to its proliferation across social networks, entertainment, immersive environments, and everyday communication (Fisher et al., 2024). Whereas deepfakes once contained clear perceptual markers of inauthenticity, such as distortions characteristic of early computer-generated contents (De Ruiter, 2021; McDonnell & Breidt, 2010), these cues are rapidly diminishing. As deepfakes reach perceptual parity with their authentic counterparts, un-

derstanding how individuals evaluate and emotionally respond to such ambiguity becomes both a theoretical and practical imperative, with implications for information integrity, social trust, and the ethical deployment of AI systems.

The prejudice against (putative) AI-generated content

Researchers have investigated the impact of authorship (human-vs. AI-generated) on perceivers' appraisal across several domains (e.g., visual art, poetry, music, faces, people). These studies typically manipulate either the *actual* origin of the stimulus (human-vs AI-generated) or the participant's *belief* about its origin, or both. Evidence has accumulated showing that participants often fail to tell apart AI-generated from human-made content at above-chance lev-

els both in artworks (Cenerini et al., 2025; Gangadharbatla, 2022; Grassini & Koivisto, 2024; Köbis & Mossink, 2021) and photorealistic images of faces created with state-of-art technology (StyleGAN3, see Stockner et al., 2026).

Interestingly, when authorship is not disclosed, AI-generated content is sometimes evaluated more positively. For instance, AI-generated poetry has been shown to receive higher appreciation ratings (e.g., beauty, lyricism, rhythm) than human-made poetry (Porter & Machery, 2024). Similarly, faces that are actually generated by AI are judged as more trustworthy than the real ones on average (Nightingale & Farid, 2022; Tucciarelli et al., 2022, study 1), possibly because generative algorithms are biased toward average, and familiar-looking faces, and familiarity breeds trustworthiness (Miller et al., 2023; but see Dunn et al., 2026).

In contrast, when stimuli are explicitly labelled as AI-generated, they are consistently evaluated more negatively than when they are presented as authentic. This “anti-AI bias” has been documented across several culturally salient domains such as visual arts (Bellaiche et al., 2023; Chiarella et al., 2022; Di Dio et al., 2023; Ragot et al., 2020), music (Ansani et al., 2025; Shank et al., 2023) and poetry (Porter & Machery, 2024). More broadly, the influence of contextual information on the evaluation of cultural products is well established and extends beyond biases specific to AI. For example, research in neuroaesthetics shows that present-

ing an artwork as an original rather than a (human-made) copy can alter both its perceived worth (Newman & Bloom, 2012), and associated neural responses (Huang et al., 2011). These findings suggest that evaluative processes are highly malleable to contextual cues and beliefs about the origin of the stimuli. The “anti-AI bias” makes no exception, inasmuch it seems driven primarily by cognitive evaluations tied to beliefs about authorship, as shown by the paradoxical appreciation of (unknowingly-) AI-made stimuli.

Interestingly, despite being strongly constrained by evolutionary pressures, more biological salient stimuli such as the perceived appearance of food (Califano & Spence, 2024) or human faces (Liefoghe et al., 2023), are also affected by the anti-AI bias. Califano and Spence (2024) report that, despite preferring AI-generated food images when unlabeled (especially ultra processed food), as soon as labels are introduced, participants tend to prefer authentic food pictures. Socially relevant stimuli like faces are also consistently rated as less trustworthy when presented as AI-generated as opposed to real (Liefoghe et al., 2023). Eiserbeck et al. (2023) reports that smiling faces labelled as AI-generated were judged less positively, processed more slowly, and elicited reduced early emotional EEG responses. On the contrary, subjective and EEG responses for angry faces were not modulated by the label, possibly reflecting the robustness of phylogenetically rooted threat stimuli. Similarly, Döring and Mohseni (2025) found that romantic images of heterosexual couples hanging out were rated more aesthetic, erotic, expressive, and vivid when labeled as photographs than as deepfakes. Taken together, these results suggest that beliefs about AI authenticity selectively dampen positive affective responses when dealing with biological salient stimuli. However, this conclusion should be interpreted with caution. In the studies discussed above, the positive stimuli are not typically comparable to the negative ones in terms of arousal, making it difficult to determine whether the observed effect truly reflects a selective dampening of positive affective responses.

One class of evolutionary salient positive stimuli that may offer a more appropriate comparison is erotic images, which are known to elicit strong motivational and attentional responses. Indeed, erotic images have been shown to capture attention as much as, or even more than, fear- and disgust-related stimuli (Ciesielski et al., 2010). In fact, they are a paradigmatic category of positively valence and highly arousing stimuli in many popular databases of stimuli for psychological research on emotion, be them visual (Kurdi et al., 2017; Lang et al., 1997; Wierzbna et al., 2015) or audio (Holz et al., 2022; Yang et al., 2018). This makes them a particularly relevant test case for examining whether contextual information, such as an AI label, can modulate responses even to highly salient positive stimuli.

 Ana Neves
 Marco Viola
 Marco Sperduti
 Guido Corradi
 Antonio Olivera-La Rosa
 Alessandro Ansani
 Marco Marini
 Dominique Makowski

Author roles were classified using the Contributor Role Taxonomy (CRediT; <https://credit.niso.org/>) as follows: Ana Neves: Data curation, Formal Analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing; Marco Viola: Conceptualization, Funding acquisition, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing; Marco Sperduti: Supervision, Writing – original draft, Writing – review & editing; Guido Corradi: Funding acquisition, Investigation, Writing – review & editing; Antonio Olivera-La Rosa: Investigation, Writing – review & editing; Alessandro Ansani: Conceptualization, Formal Analysis; Marco Marini: Conceptualization; Dominique Makowski: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing

Correspondence concerning this article should be addressed to Marco Viola, Email: marco.viola@uniroma3.it

105 AI and erotic stimuli

106 Beyond their methodological interest as biologically
107 salient stimuli for investigating the anti-AI bias, erotic stim-
108 uli are worth investigating also for their social relevance. In-
109 deed, genAI is often used to generate explicit visual con-
110 tents (Döring et al., 2024; Lapointe et al., 2025), and most
111 videos online are (non-consensual) sexual deepfakes (Ajder
112 et al., 2019; Hero, 2023). The widespread consumption of
113 AI-generated sexual content would suggest that the anti-AI
114 bias might not generalize to erotic stimuli. Or does it?

115 Easterbrook-Smith (2025) speculate that AI-generated
116 pornography might be valued less inasmuch it lacks the “aura
117 of authenticity” surrounding human-made porn. Similarly,
118 Viola and Voto (2023) propose that, inasmuch the spread
119 of deepfakes cast doubts about the authenticity of all me-
120 dia, it will end up diminishing the allure of non-consensual
121 sexual content. This hypothesis aligns with Tucciarelli et al.
122 (2022)’ findings, as they show how simply suspecting that
123 some faces within a set *could* be AI-generated lowered trust-
124 worthiness score of the whole set.

125 Yet, to the best of our knowledge, the only empirical find-
126 ings about AI-generated erotic images have been collected by
127 Zakreski et al. (2026) and Marini et al. (2024). Zakreski et
128 al. (2026) presented images of several kinds of female naked
129 bodies, including real and AI-generated pictures, reporting
130 that even if AI-generated pictures were deemed less realistic,
131 they were nonetheless considered more attractive. However,
132 the origin of the picture was not explicitly disclosed to par-
133 ticipants. Moreover, since their sample was recruited from
134 volunteers interested in sexuality and paraphilic studies in
135 Czech Republic, they alert that it “is not representative of
136 the general population” (p. 340).

137 Marini et al. (2024) performed an online experiment pre-
138 senting subjects with soft erotic images (male and female
139 models in underwear), finding that the same images were
140 rated higher in terms of sexual arousal when judged more
141 realistic (in Study 1), or presented as real as opposed to AI-
142 generated images (in Study 2).

143 However, Marini et al. (2024) studies are based on a
144 sample from a single nation (Italy) and provide scant socio-
145 demographic information. They only investigate a single de-
146 pendent variable, namely self-reported sexual arousal. How-
147 ever, self-reported first-person, *affective* feelings (“does this
148 arouses me?”) and third-person, *semantic* judgments (“is
149 that supposed to be arousing?”) have been shown to dissoci-
150 ate in some cases (Itkes & Kron, 2019). More relevantly,
151 they used only mildly sexually arousing stimuli not taken
152 from a validated dataset. Lastly, in their direct manipulation
153 of the authentic/AI-generated labeling (Study 2) they did not
154 provide any explicit manipulation check.

155 The present studies

156 The present study is a conceptual replication of Marini et
157 al. (2024), which we improved in several ways. First, we
158 employed stimuli from a validated database, the Nencki Af-
159 fective Picture System (NAPS, Marchewka et al., 2014), in-
160 cluding explicit and more sexually arousing pictures (NAPS
161 ERO, Wierzbna et al., 2015), selected via data-driven, repro-
162 ducible measures. Second, we asked participants to make
163 different evaluations, such as first-person subjective sexual
164 arousal and valence, as well as a third person (more “seman-
165 tic”) judgment on how enticing they found the image to be.

166 Sociodemographic characteristics were systematically
167 considered as a modulator of responses to stimuli, given ro-
168 bust evidence that sexual patterns differ between men and
169 women. For instance, research on arousal indicates that het-
170 erosexual men tend to exhibit greater arousal for women. In
171 contrast, heterosexual women’s responses are less gender-
172 specific (Chivers et al., 2004; Chivers, 2017; Huberman et
173 al., 2015; Safron et al., 2020).

174 Furthermore, the present studies assess whether individ-
175 ual differences in attitudes towards AI influence responses
176 to the stimuli. Beliefs about the origin of a stimulus can
177 bias evaluations, yet it remains unclear whether this effect de-
178 pends on individuals’ pre-existing attitudes towards AI. Pre-
179 vious findings suggest that such moderation is inconsistent
180 and may be domain-specific (cf. Ansani et al., 2025; Cloudy
181 et al., 2022). Moreover, some evidence suggests that broader
182 constructs such as AI readiness may attenuate anti-AI bias
183 (i.e., composite of positive AI attitudes and higher AI liter-
184 acy, Döring & Mohseni, 2025). Accordingly, the present
185 study includes attitudes towards AI as a potential modulat-
186 ing variable to test whether individual differences in these
187 attitudes shape responses to the stimuli.

188 Finally, the present studies also considers participant’s
189 sexual behaviour as modulators of responses to sexual stim-
190 uli. For example, research indicates that frequency of porno-
191 graphic consumption can influence attentional and evalua-
192 tive processing of sexual content. Repeated exposure to
193 pornographic material has been associated with attenuated
194 allocation of automatic attentional resources to sexual stim-
195 uli, suggesting a degree of habituation (Carvalho et al.,
196 2018). Furthermore, evidence suggests that individuals with
197 lower pornographic use rate erotic stimuli as less pleasant
198 compared to those with moderate or high levels of consump-
199 tion, indicating that prior exposure shapes subjective ap-
200 praisal of sexual content (Kunaharan et al., 2017). Similarly,
201 sexual activity has been shown to modulate the response
202 to erotic imagery (Prause et al., 2015; Walton & Bhullar,
203 2018; Youn, 2006). Together, these findings suggest that
204 both pornography consumption and sexual activity modulate
205 affective ratings of erotic content.

Study 1

In the first study, we aimed at gauging whether an anti-AI bias toward erotic stimuli shows up in a wide and heterogeneous sample, and whether it is modulated by demographic factors, by sexual activity or pornography consumption habits, and by attitudes toward AI. Following Marini et al. (2024), we hypothesise that images presented as AI-generated will elicit lower ratings than those presented as photographs (H1).

Additionally, in line with the literature on explicit stimuli (Huberman et al., 2015), we hypothesise that erotic images relevant to participants' sexual orientation (i.e., female images for male participants and male images for female participants) would result in higher ratings than erotic but irrelevant images and non-erotic images, and that these differences would be less pronounced for women (H2).

While prior studies have examined the effect of AI labels on affective ratings, relatively little attention has been paid to whether participants actually believe those labels, and whether belief itself modulates the observed effects. If participants second-guess the authenticity claims (e.g., suspecting that "AI-generated" images are in fact photographs, or vice versa), label-based effects would be attenuated or reversed regardless of the manipulation's objective content. Indeed, Tucciarelli et al. (2022) showed that when participants were informed that some faces in a set might be AI-generated, their responses were driven not by the objective category of each face, but by their own judgment of whether that specific face was real. Accordingly, we assess the role of individual belief in the label as an additional predictor of affective ratings (H3).

Research on general attitudes towards AI and its moderating role on ratings is mixed, with some studies finding a moderating effect (Döring & Mohseni, 2025) and others not (Liefoghe et al., 2023; Marini et al., 2024). As such, we exploratory test whether attitudes towards AI moderate responses to the stimuli (H4). Lastly, based on previous findings (e.g., Kunaharan et al., 2017), we explore whether the ratings of sexual stimuli is affected by either pornographic consumption or sexual activity (H5).

Materials and Methods

Participants

The initial sample comprised 1,067 participants recruited via multiple channels, including university research participatory schemes, social media platforms, university classrooms, and snowball sampling. This strategy yielded a heterogeneous pool of both incentivised and non-incentivised individuals, including students and members of the general population from England, France, Italy, Colombia, and Spain. Data collection occurred from the 19th of January to 1st of June 2024. At this time, publicly available image

generators such as OpenAI's DALL-E 3 (Betker et al., 2023) were producing highly photorealistic outputs.

Eight participants were removed if they showed no variation in arousal ratings across trials, and four because they displayed a salient negative correlation between arousal and enticement alongside lower arousal ratings for erotic compared to neutral stimuli, suggesting a possible misunderstanding of response scales direction. In order to preserve homogeneous and well-powered samples, participants identifying with a gender other than Female or Male ($N = 15$) were excluded from further analysis, as were female and male participants reporting a sexual orientation other than Heterosexual ($N = 335$). The latter comprised 150 bisexual females, 72 bisexual males, 32 homosexual females, 38 homosexual males, and 43 participants (23 females, 20 males) who reported another orientation. The final sample consisted of 705 heterosexual participants (Mean age = 30.2 ± 11.8 years; 35.7% female). Participants were primarily from the United Kingdom (28.23%), Italy (18.72%), the United States (14.33%), and Colombia (11.06%), with the remaining 27.66% distributed across other countries. Ethical approval for this study was obtained from the University of Sussex Ethics Committee (ER/MHHE20/1).

Procedure

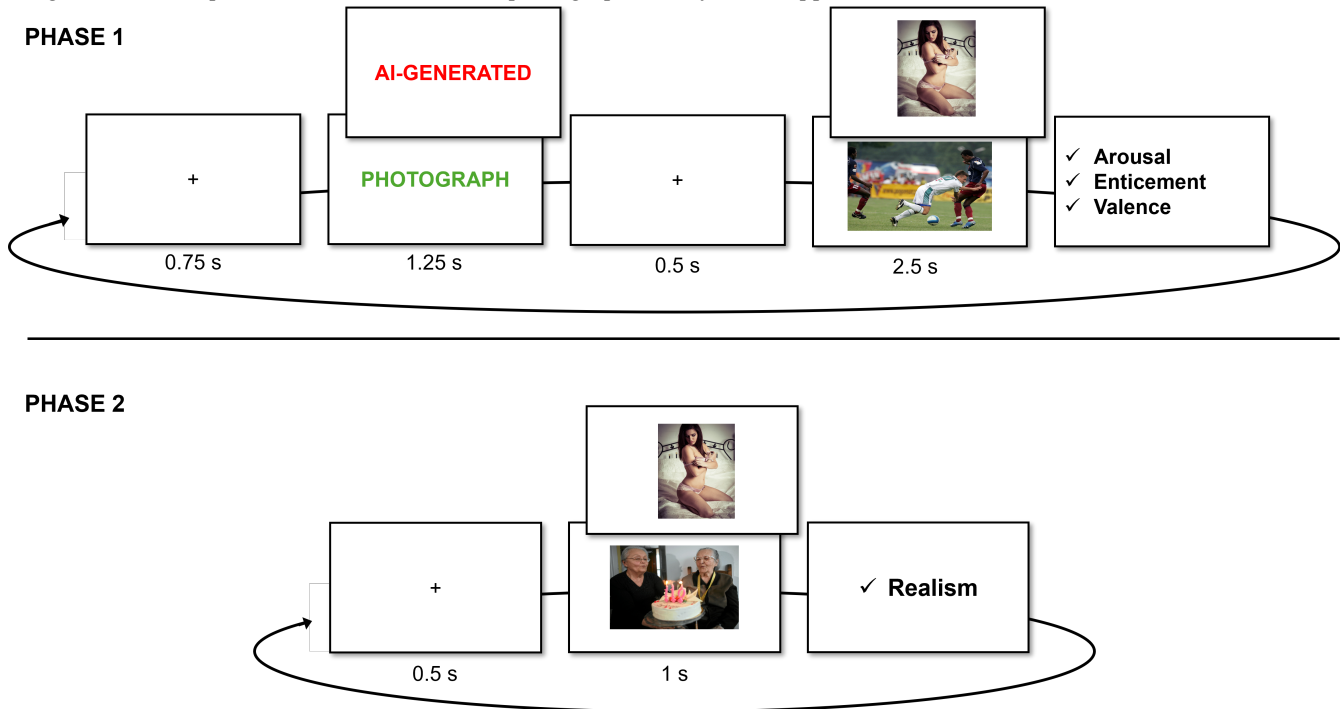
The study was conducted in line with the born-open principle (De Leeuw, 2024), ensuring transparency and reproducibility at every stage. The experiment was implemented with *JsPsych* (De Leeuw, 2015) and all materials, including its code, the items, instructions and exact procedure, is available in open-access at <https://github.com/RealityBending/FictionEro/tree/main/study1/experiment>. Note that the experiment was translated into English, French, Italian and Spanish (based on our native language) and then we disseminated it in the relevant countries.

Phase 1: Affective Evaluation. Participants were first presented with a cover story introducing the goal of the study as a validation of a new image-generation algorithm trained to produce high-quality erotic (and also non-erotic) content. Participants were informed that they would be shown a series of images, some of which were generated by the algorithm and others that were real photographs, and that their task was to rate each image on several affective dimensions, each preceded with a cue (the word "AI-Generated" or "Photo") presented for 1.25 s.

Participants' task was to rate each image on three dimensions on analog scales: *Arousal* ("How much do you find the image sexually arousing? This question is about your own personal reaction felt in your body when seeing the image"), *Enticement* ("How enticing and sexually appealing would you rate this image to be. Think of how much, in general, people similar to you in terms of gender and sexual orientation would like it"), and *Valence* ("Did the image evoke a

Figure 1

Overview of the experimental procedure for Study 1. A) Phase 1: Participants viewed a randomized sequence of erotic and non-erotic images. After each image, they provided ratings on three dimensions: sexual arousal, enticement, and affective valence. B) Phase 2: The same set of images was presented again in a new random order. This time, participants rated each image based on its perceived realism (i.e., how photographic or lifelike it appeared).



positive and pleasant (not necessarily sexual) feeling in you, or could it be better characterized as negative and unpleasant? Think of how much you did enjoy (or not) looking at the images”).

Unbeknownst to participants, all 60 stimuli were real images (presented for 2.5 s), and the cues were randomly assigned. We selected 60 images: 40 erotic images from the NAPS ERO (Wierzbica et al., 2015) and 20 non-erotic images from the original NAPS database (Marchewka et al., 2014), using the NAPS ERO normative ratings from heterosexual men and women for all images. The erotic images formed two sets of 20, one selected on men’s arousal norms and one on women’s. Each set comprised, within each image category (male, female), the eight highest- and two lowest-arousal images (i.e., 10 male and 10 female images per set), with the women-normed set drawn from images not already selected for the men-normed set. Non-erotic images (NAPS “Faces” and “People” categories) were selected on arousal averaged across men’s and women’s ratings: the 10 lowest-arousal (neutral) images, and the 10 highest-arousal images among those with mean valence above the scale midpoint (positively arousing).

Questionnaires. The *Beliefs about Artificial Image Technology* (BAIT) questionnaire (first used in Makowski, Te, et

al., 2025) was refined to include, additionally to 6 items from the GAAIS (Schepman & Rodway, 2020, 2023) measuring negative and positive attitudes towards AI, 8 items specifically developed to assess beliefs about computer-generated material across various types such as text, videos, images, and VR-environments. As such, the BAIT includes two dimensions: expectations ($\omega = .70$) and attitudes ($\omega = .72$) both with acceptable internal consistency. Additionally, the first stand-alone item assesses self-reported AI knowledge on a 6-point Likert scale (Not at all - Expert).

Two items from the *Consumption of Pornography Scale* (COPS, Hatch et al., 2023) were included to assess pornography use frequency and duration. An additional item assessed the recency of any sexual activity, with response options ranging from within the past 24 hours to more than a year ago.

Phase 2: Realism Beliefs. Following the questionnaires, participants were presented with the instructions of phase 2, assessing the perceived realistic quality of the stimuli. The instructions were framed as follows: “In the next phase, we would like to see if you found our image generation algorithm convincing and error-free. We will briefly present you all the images one last time (the AI-generated ones, as well as the photos), and you will have to rate them on how real

(how realistic, photography-like) the image is". Participants then viewed the same 60 images (1 s) in a newly randomised order and rated the stimuli on realism with an analog slider scale (AI-generated - Photograph).

Following phase 2, a manipulation check was done by offering participants the opportunity to tick boxes pertaining to their experience, including items prompting whether they could distinguish which images were AI-generated or perceived no difference between photographs and AI-generated images; whether they felt the AI-generated images were more or less arousing; whether they believed the labels were inaccurate or reversed; and whether they found some images arousing or none at all. Finally, participants were debriefed and were informed of the true purpose of the study, and of the fact that all images were actually genuine pictures.

Data Analysis

Bayesian Mixed Models were fitted separately for Arousal, Valence, and Enticement as dependent variables, with Gender (Female vs. Male), Condition (AI-Generated vs. Photograph), Relevance (Relevant, Irrelevant, Non-Erotic), and Congruence (True vs. False). Relevance corresponded to the sexual interest according to the participants' gender and sexual orientation (i.e., erotic pictures involving females were categorized as relevant for heterosexual males, and irrelevant for heterosexual females, and vice versa for pictures of males). Congruence, a proxy of condition belief, was also computed based on whether the condition matched the (dichotomized) reality rating of phase 2.

As the outcome variables (measured on analog sliders) showed excess responses at the bounds, Zero-one inflated beta (ZOIB) models (Heiss, 2021; Vuorre, 2019) were used. Participants and items were entered as random intercepts, with the effect of relevance and condition also modeled as variable within participants. Conservative priors ($Normal(0, 0.5)$) were placed on all fixed effects. The models were run using the brms package (Bürkner, 2017) and analysed using the easystats collection of packages (Lüdtke et al., 2021, 2022; Makowski et al., 2019).

For all outcomes, we assessed the difference between the conditions (Photograph - AI-Generated) using marginal contrasts (Makowski, Ben-Shachar, et al., 2025), stratified by relevance, gender, and congruency. Coefficients are considered as significant when the 95% credible interval (CrI) is strictly non-null. As the patterns of results for the outcomes were similar, we will focus on presenting the detailed results for Arousal as our primary outcome, and highlighting when divergent trends are present for Enticement or Valence.

To evaluate the reliability of individual-level parameters (i.e., random intercepts and random slopes for each participant and each item), the Variance-over-Uncertainty Ratio index (D-vour, Lüdtke et al., 2021) will be computed. This index estimates the reliability of group-level variance by

quantifying the normalised ratio of observed between-group variability to the uncertainty associated with the corresponding random-effect estimates. Values of D-vour greater than 0.666 indicate moderately reliable random-effect estimates, corresponding to a 2:1 ratio of between-group variance to estimation uncertainty. In this context, reliability refers to cases in which variability between groups (e.g., participants) exceeds the uncertainty in the estimates, suggesting that the parameter can meaningfully capture inter-individual differences.

Lastly, a correlation matrix will be computed between the models' individual-level estimates and potential moderators: beliefs, attitudes towards and knowledge of AI, porn frequency and recency of sexual activity. Correlations, computed with the correlations package (Makowski et al., 2022) will be used to explore whether reliable variation in the random effects can be explained by theoretically relevant individual differences. Pearson's r and its associated 95% confidence interval (CI) will be reported and interpreted according to Pearson's criteria (Cohen, 2013). All analyses were conducted in R (version 4.5.2). Anonymized data, together with all pre-processing and analysis scripts, will be openly released on <https://github.com/RealityBending/FictionEro/tree/main/study1/> to facilitate complete reproducibility.

Results

Manipulation Checks

An initial manipulation check was conducted to assess participants' belief in the instructional cues. Based on participants' feedback after the final rating phase, 41.7% of participants reported that they did not believe the labels, indicating that a large proportion of participants perceived the image classifications as incorrect.

To examine whether participants' explicit beliefs about AI-generated content corresponded to their affective responses, Bayesian independent-samples t-tests compared the difference scores between images labelled as AI-generated and as photographs as a function of whether participants reported finding AI images less arousing.

For arousal, participants who reported that AI images were less arousing showed a larger difference between images labelled as AI-generated and as photographs ($M_{\text{difference}} = -0.02$) than participants who did not report this belief ($M_{\text{difference}} \approx 0$). A Bayesian independent-samples t-test indicated moderate evidence for a group difference ($BF_{10} = 3.45$), with a small-to-moderate standardised effect size (median $\delta = 0.30$, 95% CrI [0.07, 0.54], $pd = 99.5\%$). A similar pattern was observed for enticement and valence (for full results see the analysis folder on the GitHub link above).

Marginal Contrasts

Effect of Relevance

To evaluate whether the different stimulus types (female, male and non-erotic images) elicit dissimilar responses, marginal contrasts from the mixed models were examined, focusing on the effect of relevance. These contrasts assessed differences in arousal ratings across stimulus types separately for males and females.

For males, female erotic images were rated as more arousing than both male erotic ($\Delta = 0.43$, 95% *CrI* [0.39, 0.46], $pd = 100\%$) and non-erotic images ($\Delta = 0.44$, 95% *CrI* [0.40, 0.47], $pd = 100\%$). For females, male erotic images were rated as more arousing than non-erotic images ($\Delta = 0.20$, 95% *CrI* [0.17, 0.24], $pd = 100\%$), but not credibly more than female erotic images ($\Delta = 0.04$, 95% *CrI* [0.00, 0.08], $pd = 96.36\%$). Similar patterns were found for enticement and valence; although for females non-erotic images were rated as more positively valenced than male erotic images.

Effect of Condition

Estimated marginal contrasts assessed the effect of condition (photographic vs. AI-generated) across gender and relevance. For males, female erotic images labelled as photographs were rated as slightly more arousing than those labelled as AI-generated, although the effect was small ($\Delta = 0.02$, 95% *CrI* [0.00, 0.03], $pd = 99.6\%$). Females showed the opposite pattern, rating male erotic images labelled as AI-generated marginally higher than those labelled as photographs ($\Delta = -0.02$, 95% *CrI* [-0.03, 0.00], $pd = 98.3\%$). No other contrasts provided credible evidence for differences (all $pd < 88.2\%$). Similar results were seen for enticement and valence ratings for males, but no credible differences between images labelled as photographs or AI-generated for females.

Effect of Condition Depending on Congruence

Estimated marginal contrasts examined the effect of condition as a function of gender, relevance, and congruence (see Figure 2). In congruent trials, males rated female erotic images labelled as photographs as more arousing than those labelled as AI-generated ($\Delta = 0.08$, 95% *CrI* [0.06, 0.10], $pd = 100\%$), and females showed the same pattern for male erotic images, with a slightly smaller effect ($\Delta = 0.06$, 95% *CrI* [0.04, 0.08], $pd = 100\%$).

In incongruent trials, this pattern reversed for the same stimulus types: images labelled as AI-generated were rated as more arousing than those labelled as photographs. This reversal was more pronounced in females ($\Delta = -0.09$, 95% *CrI* [-0.12, -0.07], $pd = 100\%$) than in males ($\Delta = -0.05$, 95% *CrI* [-0.07, -0.03], $pd = 100\%$). Similar crossover patterns were observed for enticement and valence ratings of erotic, relevant images.

For the remaining stimulus types (male erotic and non-erotic images for male participants; female erotic and non-erotic images for female participants), differences between the label conditions were generally small, and several credible intervals included zero. In congruent trials, males rated non-erotic images labelled as AI-generated as slightly less arousing than those labelled as photographs ($\Delta = 0.02$, 95% *CrI* [0.01, 0.04], $pd = 99.66\%$), and a comparable effect was observed in females for female erotic images ($\Delta = 0.02$, 95% *CrI* [0.01, 0.04], $pd = 100\%$). In incongruent trials, this pattern reversed; however, a credible difference emerged only for females' ratings of female erotic images, which were higher when labelled as AI-generated ($\Delta = -0.03$, 95% *CrI* [-0.05, -0.01], $pd = 99.69\%$). For enticement ratings the same reversal pattern was found, with credible differences for all with the exception of males' ratings of non-erotic images. On the other hand, all differences were credible for valence ratings for both genders and the same crossover effect was found.

Discussion

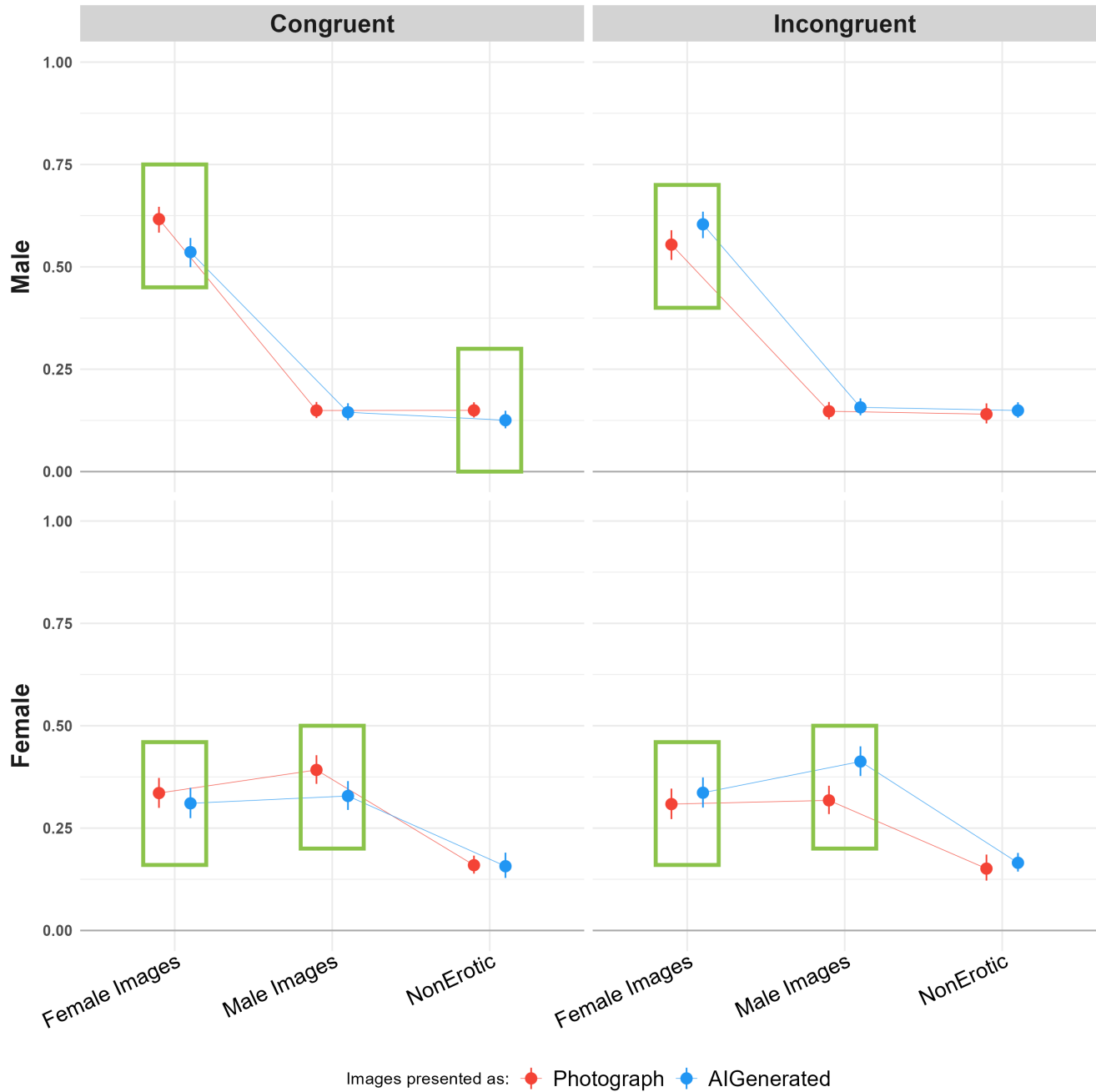
In our first study, we presented subjects (heterosexual males and females) with several images from the NAPS database, including non-erotic images and both irrelevant (same gender) and relevant (other gender) erotic images. To manipulate subjects' belief about the authenticity of the stimuli, each image was cued with a colored label presenting it as either "photograph" or "AI-generated". After collecting subjective ratings of arousal, enticement and valence for each image, we performed a manipulation check to assess the efficacy of our experimental manipulation by asking them to rate how realistic they deemed the image.

In partial alignment with our H1 and with Marini et al. (2024), for male participants, images labelled as AI-generated were rated as slightly less arousing than those labelled as photographs. For females, however, this pattern reversed. In both cases the effect size was small. No credible differences between conditions emerged for irrelevant or non-erotic images, suggesting this gendered divergence was specific to sexually relevant content rather than a general labelling effect.

As per H2, marginal contrasts indicated that males, but not females, rated relevant erotic images as more arousing than irrelevant erotic images. This is consistent with prior literature demonstrating stronger category-specificity in men's subjective responses to explicit stimuli (Chivers et al., 2004; Chivers, 2017; Huberman et al., 2015). Prior work suggests that women's sexual responses are more sensitive to contextual and relational features, such as narrative, emotional connection, and cues of intimacy (Laan et al., 1994; Rupp & Wallen, 2007, 2009). The stimuli in the present study did not include depictions of couples, as the limitations of genAI at the time could have made them suspicious.

Figure 2

Estimated means for the effect of Condition depending on Congruency. Green rectangles represent differences in means where the $pd > 97.5\%$, indicating a consistent effect direction (Makowski et al., 2019). A crossover effect can be seen where the difference between arousal ratings between images labelled as AI-generated and those labelled as photographs is reversed for incongruent trials. Of note, female images are relevant for heterosexual males while irrelevant for heterosexual females.



Notably, the label promoted a minimal difference in ratings for both genders. To further examine the role of stimulus evaluation, as per H3, we included congruency between the label presented in Phase 1 and participant's realism ratings in Phase 2 as a predictor. A crossover interaction emerged: when participants' Phase 2 ratings were incongruent with the initial label, AI-labelled erotic images were rated as more arousing; conversely, when ratings were congruent, images labelled as photographs were rated as more arousing across both genders.

We also tested whether variability in responses could be explained by individual differences in participants' attitudes towards AI (H4) and their sexual behaviour (H5). To do so, we first assessed the extent to which responses to the AI label varied meaningfully between participants. We did not find reliable correlations between participants' features and the effect of how the label influenced their responses, possibly due to the fact that between-participants variability in the sensitivity to the AI label was minimal (full details are available in the analysis script on GitHub). One possible explanation is that the label manipulation was not sufficiently strong or salient to elicit differentiated responses across individuals.

Study 2

At the time of Study 1, limitations in genAI realism prevented convincing depictions of couples having sexual intercourse, so the role of relational context could not be tested directly. Recent advances now make the (mis)labeling of realistic couple photographs as AI stimuli credible, making this test possible. Given that (heterosexual) women's arousal tends to be more sensitive to intimacy, narrative, and the presence of an interacting partner than men's (Laan et al., 1994; Rupp & Wallen, 2007, 2009), we included couple-based stimuli in Study 2 to assess whether the absence of relational context contributed to the lower arousal ratings observed among female participants in Study 1.

Second, in Study 1, marginal contrasts indicated credible differences in ratings between images labelled as AI-generated and those labelled as photographs only for relevant erotic stimuli, in both genders. Study 2 therefore focused exclusively on erotic images relevant to participants' gender and sexual orientation.

Third, Study 1 demonstrated that the effect of AI labelling was small and showed minimal variability across participants, limiting our ability to examine individual differences. Yet, given that this effect was greater when the manipulation was believed (as per our H3) to strengthen the effectiveness of the manipulation, Study 2 used a more elaborate cover story. Moreover, the control task was more explicit, requiring the participants to commit to a judgement of whether each image was an actual photograph or AI-generated.

Based on these aims, we tested the following hypotheses. As in Study 1, we hypothesised that images labelled as AI-

generated would be rated lower in arousal than those labelled as photographs (H1). We also hypothesised that the inclusion of couple-based stimuli would narrow the gender gap in arousal observed in Study 1 (H2). Given the crossover effect observed in Study 1, we hypothesised that this pattern would be stronger for congruent trials, and reverse for incongruent ones (H3). Finally, as in the previous study, we tested whether attitudes towards AI (H4) and sexual activity and pornographic consumption (H5) moderate responses to the stimuli.

Materials and Methods

Participants

The initial sample comprised 279 participants recruited via Prolific®. Inclusion criteria required participants to be native English speakers or residents of countries with high levels of English proficiency. Data collection occurred from 27 of January 2025 to 31 of July 2025. By this time, publicly available image generators were producing outputs that people could no longer reliably distinguish from authentic photographs (Cooke et al., 2025).

Participant exclusions were applied as follows: five participants were removed for showing no variability in arousal ratings (i.e., they did not move the response scales). An additional five participants were excluded for completing the study on a mobile device. One participant was excluded due to displaying negative correlations between arousal and both enticement and valence. Furthermore, five participants who self-identified as neither female nor male, and two participants who reported a sexual orientation other than heterosexual, homosexual, or bisexual, were excluded from further analyses due to lack of sufficient group sizes. Finally, one participant was removed because the stimuli presented were not relevant to their gender and sexual orientation.

The final sample consisted of 261 heterosexual participants (Mean age = 37.4 ± 12.7 , 48.7% Female). 56.32% of participants were from the United Kingdom, 26.82% from South Africa, 10.34% from the United States, and the remaining 6.51% were from other countries.

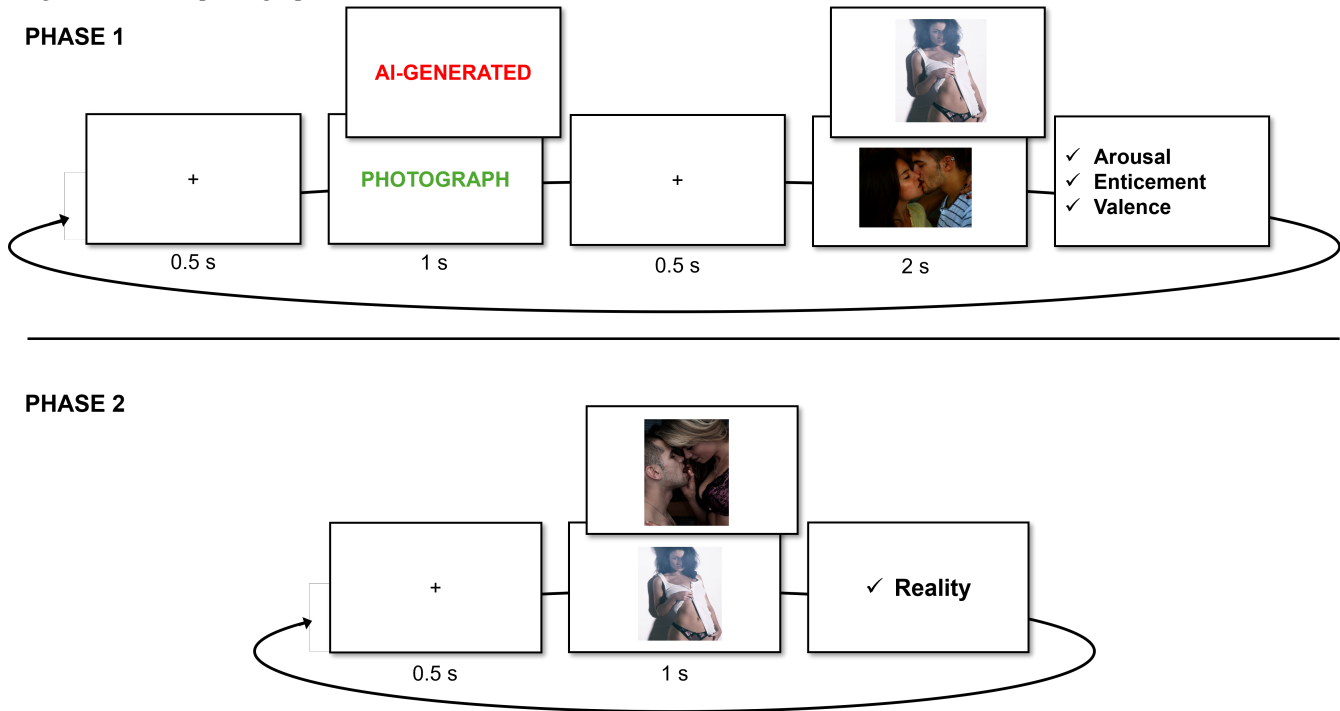
Ethical approval for this study was obtained from the School of Psychology Ethics Committee at the University of Sussex (ER/EB672/2).

Procedure

Consistent with Study 1, Study 2 was conducted in jsPsych following born-open principles (De Leeuw, 2024) and implemented in *JsPsych* (De Leeuw, 2015) and all data, materials, including its code, the items, instructions and exact procedure, is available in open-access at <https://github.com/RealityBending/FictionEro/tree/main/study2/experiment>

Figure 3

Overview of the experimental procedure for Study 2. A) Phase 1: Participants viewed erotic images in randomised order. Each image was preceded by a randomised label and followed by ratings of sexual arousal, enticement, and valence. B) Phase 2: The same images were presented in a new randomised order, and participants rated their confidence that each image was AI-generated or a photograph.



Phase 1: Affective Evaluation. In this study, participants were informed that the researchers were collaborating with a young AI start-up based in Brighton, UK, specializing in making more ethical AI content. This was intended to enhance the believability of the study. Participants were then informed that they would be shown a series of highly realistic AI-generated erotic images intermixed with real photographs, each preceded by a cue (as in Study 1) presented for 1s.

The *Arousal*, *Enticement* and *Valence* ratings were defined as per Study 1 (see above) but this time they were rated on a 7-point Likert scale from Not at all (0) to Very much (6).

As in Study 1, all stimuli were real photographs (presented for 2 s) from the NAPS ERO (Wierzbna et al., 2015) and the labels were randomly assigned. Unlike Study 1, no non-erotic images were included. Each participant viewed 50 images relevant to their gender and sexual orientation: 25 depicting individuals and 25 depicting opposite-sex couples. For the female-individual and male-individual categories, all 25 available NAPS ERO images were used. For opposite-sex couples, we selected the 25 (of 50 available) with the highest arousal summed across heterosexual men's and women's

normative ratings; among three images tied at the selection threshold, the one with the lowest summed valence was excluded.

After viewing all images, participants completed a brief feedback asking them to indicate whether they found some or no images particularly arousing, and whether AI-generated images were more or less arousing than the photographs. Participants were also asked whether differences between AI-generated and real images were obvious or subtle, whether they perceived any inconsistencies or reversals in the labeling, and whether they believed all images were either photos or AI-generated. If participants indicated that they believed all images were either real or AI-generated, they were further asked to rate their confidence in this judgment on a scale from "Not at all" to "Completely certain".

Questionnaires. The questionnaires used in Study 2 were largely the same as those in Study 1, with minor modifications. In the BAIT scale, two items, assessing beliefs that AI might take control of people and interest in using AI systems in daily life, were removed. Three items assessing AI expertise were added (e.g., "I consider myself an expert in AI technology"), as well as four items assessing one's ability to discriminate between AI and real stimuli (e.g., "I can easily distinguish between real and AI-generated images").

Lastly, 7 items assessing AI-bias were further included (e.g., “Human creators bring a unique perspective that AI cannot replicate”). This version of the BAIT resulted in 5 dimensions: Expectations ($\omega = .76$), Attitudes ($\omega = .78$), Expertise ($\omega = .77$), Discrimination ($\omega = .80$) and Bias ($\omega = .77$). Items were rated in a 7 Likert scale from 0 (Disagree) to 6 (Agree). Additionally, the wording was streamlined by replacing “Artificial Intelligence” with “AI” throughout the scale.

In the COPS, the item assessing the typical duration of pornography viewing sessions was omitted, retaining only items measuring frequency of pornography viewing and recency of sexual activity.

Phase 2: Reality Beliefs. Phase 2 replicated that of Study 1, presenting participants with all images again (for 1 s) in a randomised order and assessing the perceived authenticity of the stimuli. The instructions, however, made the manipulation more overt by explicitly disclosing the mislabeling: “But there is something important we need to tell you... In the previous phase, some images were intentionally mislabeled (we told you it was a photograph when it was actually AI-generated, and vice versa).” Participants were then asked to judge the true category of each image by indicating their confidence (i.e., “Indicate your confidence that the image is fake or real”) on a continuous scale from AI-Generated (−3) to Photograph (3).

Following completion of phase 2, participants provided general feedback regarding their experience and any additional comments. Finally, participants were presented with a debrief page informing them that all images were, in fact, real photographs.

Data Analysis

As in Study 1, Bayesian linear mixed models will be fitted for the three outcome variables, each predicted from Gender (Female vs. Male), Condition (AI-generated vs. Photograph), and Congruence (True vs. False). For ease of comparison with study 1 results, all outcome variables will be bounded between 0 to 1, and the ZOIB family will be fitted. Conservative priors ($Normal(0, 0.5)$) will be placed again on all fixed effects.

Marginal contrasts will be computed to compare marginal means with condition (i.e., AI-generated and Photograph) as the predicted contrasts, stratified by gender and congruency. The same individual-level parameter and moderator-correlation analyses as in Study 1 will be conducted, using the same packages. To further assess the effect of the type of image (e.g., couple vs individuals) the same analysis will be conducted adding the type of image (image of individuals vs image of couples) as predictor.

Consistent with Study 1, the Results section will focus on Arousal as the primary outcome, highlighting where divergent trends emerge for enticement or valence; full results for all outcomes are available in the analysis scripts on GitHub.

All analyses were conducted in R (version 4.5.2), and all materials, data, and analysis scripts for Study 2 are available at <https://github.com/RealityBending/FictionEro/tree/main/study2>.

Results

Manipulation checks

In Study 2, 32% of participants reported believing that all images were real based on their feedback after Phase 1, indicating that a third of participants did not believe in the manipulation.

To examine whether participants’ explicit beliefs about AI-generated content corresponded to their affective responses, Bayesian independent-samples t-tests compared the difference scores between images labelled as AI-generated and as photographs as a function of whether participants reported finding AI images less arousing. For arousal, participants who reported finding AI images less arousing showed a numerically larger difference between images labelled as AI-generated and as photographs ($M_{\text{difference}} = -0.03$) than those who did not ($M_{\text{difference}} = -0.009$), but there was no credible evidence for a group difference ($BF_{10} = 0.47$; median $\delta = 0.26$, 95% *CrI* [−0.17, 0.69], $pd = 88.6\%$). Similarly, no credible group differences emerged for enticement or valence.

Marginal contrasts

Effect of Type

The addition of image type (i.e., images of couples vs. images of individuals) revealed a credible gender difference in ratings of individual images, with males rating them higher in arousal than females ($\Delta = 0.17$, 95% *CrI* [0.11, 0.22], $pd = 100\%$). No credible differences between males and females were found for ratings of couple images ($\Delta = 0.009$, 95% *CrI* [−0.02, 0.04], $pd = 69.92\%$). The same patterns were found for enticement and valence (see Figure 4).

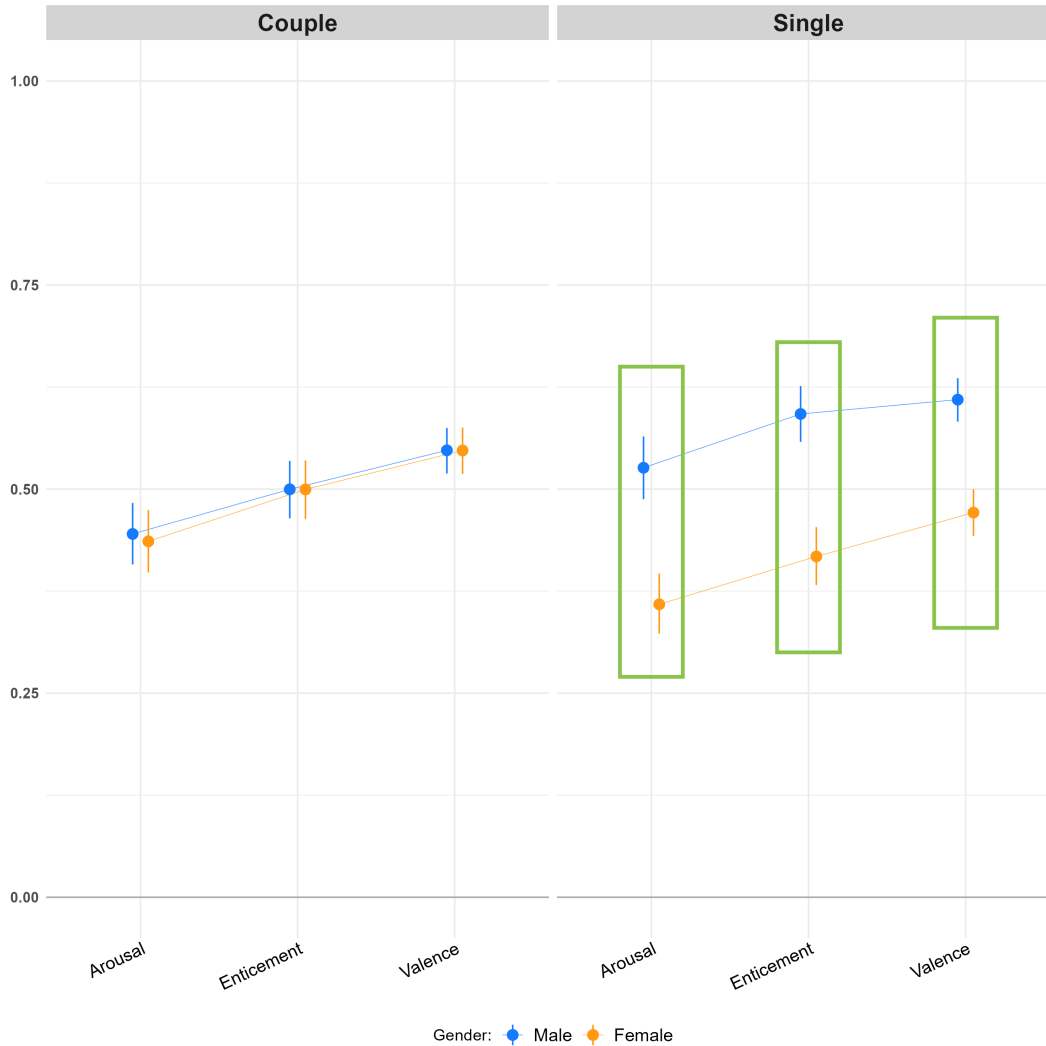
Additionally, credible differences between image types were found within each gender. Females rated couple images higher in arousal than images of individuals ($\Delta = 0.08$, 95% *CrI* [0.03, 0.12], $pd = 99.98\%$), whereas males rated couple images lower ($\Delta = -0.08$, 95% *CrI* [−0.12, −0.04], $pd = 99.98\%$). Enticement and valence ratings were also credibly different for both genders ($pd > 99.99\%$), with males rating couple images lower than images of individuals, and females showing the reversed pattern.

Effect of Condition (with and without Type)

Estimated marginal contrasts assessing the effect of condition across gender showed that the difference was only credible for Males ($\Delta = 0.06$, 95% *CrI* [0.04, 0.07], $pd = 100\%$) but not for females ($\Delta = 0.01$, 95% *CrI* [0.00, 0.02], $pd = 93.57\%$). The same pattern was found for enticement. For

Figure 4

Estimated means for Arousal, Enticement and Valence for images of individuals and couples, for both genders. Green rectangles represent significant differences.



valence ratings, both genders rated images labelled as photograph higher than those labelled AI-generated ($pd > 98.8\%$).

The effect of condition across gender for the type of images showed only males credibly differed in arousal ratings of images of couples labelled photograph vs those labelled AI-generated ($\Delta = 0.02$, 95% CrI [0.00, 0.04], $pd = 99.23\%$), and ratings of individuals ($\Delta = 0.02$, 95% CrI [0.00, 0.03], $pd = 98.77\%$). No credible differences were found for females ($pd < 88.2\%$). Enticement ratings followed the same pattern, while for valence ratings the only credible difference was found for females ratings of couple images, with those images labelled photograph being rated higher than those labelled AI-generated ($\Delta = 0.02$, 95% CrI [0.01, 0.04], $pd = 99.52\%$).

Effect of Condition Depending on Congruence (with and without Type)

Estimated marginal contrasts examined differences in arousal ratings between images labeled as photographs or AI-generated as a function of gender and congruence (see Figure 5). Under congruent trials, images labelled as photographs were rated as more arousing for both genders, with a stronger effect for males ($\Delta = 0.06$, 95% CrI [0.04, 0.08], $pd = 100\%$) than females ($\Delta = 0.02$, 95% CrI [0.00, 0.04], $pd = 97.18\%$). Under incongruent trials, the direction of the effect reversed for males ($\Delta = -0.02$, 95% CrI [-0.04, -0.01], $pd = 99.71\%$), whereas the effect for females was negligible ($\Delta \approx 0$, 95% CrI [-0.02, 0.02], $pd = 57\%$).

Incorporating image type into the models provided lim-

ited additional evidence for differences overall, although some credible effects emerged for males. Specifically, under congruent trials, males showed higher arousal ratings for photograph- relative to AI-labelled images for both couples ($\Delta = 0.06$, 95% *CrI* [0.03, 0.08], *pd* = 100%) and individuals ($\Delta = 0.06$, 95% *CrI* [0.04, 0.08], *pd* = 100%). Under incongruent trials, a credible difference for males was observed only for individual images ($\Delta = -0.03$, 95% *CrI* [-0.05, -0.01], *pd* = 99.41%). In contrast, female participants showed no credible differences across trial types or image categories, with estimates centred near zero and credibility intervals spanning zero.

Enticement and valence ratings followed a broadly similar pattern to arousal, with higher ratings for photographic images under congruent trials and weaker, less consistent reversals under incongruent trials. A notable exception was observed for female participants, who showed credible differences under congruent trials, with photograph-labelled images rated higher for both enticement and valence (*pd* > 99.2%).

When image type was included in the models, credible differences in enticement ratings were largely restricted to males: under congruent trials, males showed higher ratings for photograph-labelled images for both couples and individual stimuli (*pd* = 100%), whereas under incongruent trials, this effect remained only for individual images (*pd* > 99%). For valence ratings, most contrasts became credible following the inclusion of image type (*pd* > 98%), maintaining the same reversal pattern under incongruent trials. The primary exception was for female ratings of couple images under incongruent trials, where evidence remained inconclusive (*pd* < 59.5%).

Discussion

In Study 2, participants viewed erotic images depicting both couples and individuals matched to their sexual orientation. As in Study 1, images were labelled as either photographs or AI-generated. Following Phase 1, 32% of participants reported believing that all images were real. Although this measure differs from Study 1's manipulation check (where 41.7% reported disbelieving the labels in any form) and the two figures are therefore not directly comparable, the pattern is at least consistent with the strengthened cover story in Study 2 having improved the manipulation's credibility.

Overall, the pattern of findings partially replicated those observed in Study 1. Considered independently of congruence, the AI-labelling effect on arousal was credible for males but not for females, providing support for H1 in males only.

A key aim of Study 2 was to examine whether the inclusion of relational context (i.e., via couple-based stimuli) would narrow the gender gap in arousal observed in Study 1

(H2). The findings partially support this prediction. Specifically, males and females did not differ credibly in their ratings of couple images, whereas males continued to rate individual images as more arousing than females. However, this convergence did not arise solely from the predicted mechanism of selectively increased female arousal: females rated couple images higher in arousal than images of individuals, but males rated couple images credibly lower than images of individuals. The gender gap therefore appears to have narrowed through change in both directions (i.e., increased arousal to couples among females and decreased arousal to couples among males) rather than through a female-specific increase.

As per H3, under congruent trials, images labelled as photographs were rated as more arousing, enticing, and positively valenced compared to those labelled as AI-generated. Under incongruent trials, this pattern reversed for males, consistent with the crossover effect observed in Study 1. For females, however, no credible reversal emerged under incongruent trials, indicating an absence of the predicted effect rather than merely a weaker version of it; some credible differences did emerge for females under congruent trials for enticement and valence ratings. Thus, while the predicted crossover interaction was observed, support was essentially confined to male participants, only partially replicating Study 1.

The overall effect of AI labelling remained small and showed minimal reliable variability across participants, replicating the pattern observed in Study 1. Consequently, as in Study 1, it was not possible to draw strong conclusions regarding individual differences as moderators of the labelling effect (H4-H5). Weak associations were found between arousal responses (both baseline and AI-related parameters) and self-reported beliefs, attitudes, and AI expertise (see full analysis on the link provided above).

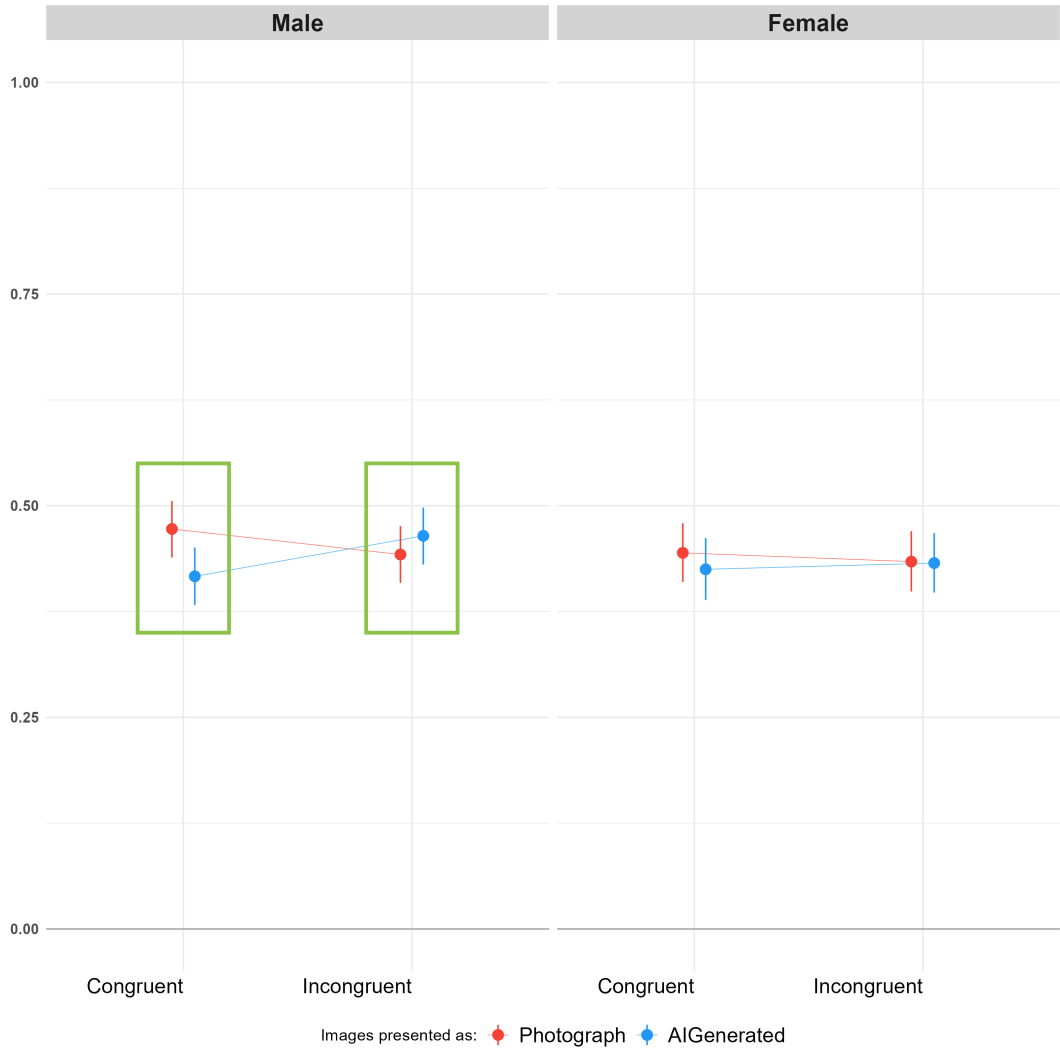
General Discussion

The present paper investigated the anti-AI bias in response to erotic stimuli across two studies. Overall, the findings provide converging evidence for three main conclusions: (1) there is a consistent but small effect of AI labelling on responses that is primarily observed in males, (2) there is a strong gender difference in response to erotic stimuli, that is robust and context dependent, and (3) belief-related processes (i.e., congruency between initial labels and subsequent realism/reality judgements) play a central role in shaping affective responses.

In line with previous findings (e.g., [Marini et al., 2024](#)), images labelled as AI-generated were generally rated as less arousing than those labelled as photographs. However, across both studies this effect was small in magnitude and primarily driven by male participants, whose ratings were higher in arousal, enticement, and valence, for images la-

Figure 5

Estimated arousal means for the effect of Condition depending on Congruency. Green rectangles represent significant differences.



beled as photographs than for those labeled as AI-generated. On the contrary, in Study 1, females rated relevant erotic images higher in arousal (but not valence or enticement) when they were presented as AI-generated. This reverse pattern was not found in Study 2.

Our results also replicate and extend prior literature on gender differences in response to erotic stimuli (Chivers et al., 2004; Chivers, 2017; Huberman et al., 2015; Safron et al., 2020). In Study 1, males rated sexually relevant images as more arousing than irrelevant ones, whereas females did not. Study 2, which presented only relevant stimuli, examined whether the gender gap in arousal observed in Study 1 was contingent on the nature of the stimuli. When relational context was introduced through couple-based images, this

gap narrowed substantially: males and females did not differ in their ratings of couple images, while a gap persisted for images depicting individuals. Notably, Study 2 also showed that females rated images of couples higher than images of individuals, while males did the opposite. This finding supports the interpretation that female sexual responses are more sensitive to contextual and relational cues (Laan et al., 1994; Rupp & Wallen, 2007, 2009) and suggests that the absence of such features in Study 1 likely amplified the apparent gender gap in affective reactions.

Third, and most importantly, the present findings highlight the central role of belief-related processes in shaping affective responses. In both studies, when asked for feedback, a substantial proportion of participants expressed doubts

about the veracity of the labels. We further evaluated these belief-related effects on ratings by operationalising a congruency variable, which revealed a crossover interaction with condition. When participants' realism judgments were congruent with the initial label, images labelled as photographs were rated higher than those labelled as AI-generated. In contrast, when judgments were incongruent, this pattern reversed, with AI-labelled images receiving higher ratings, suggesting that participants' affective responses are closely tied to whether they were second-guessing the authenticity of the images.

We had additionally hypothesised that individual differences in AI attitudes and beliefs, pornography consumption frequency and recency of sexual activity might be associated with the strength of this labelling effect. However, individual-level estimates of sensitivity to the AI-generated label showed minimal reliable variability between participants, meaning these hypotheses could not be meaningfully tested in the present data; weak correlations with these variables should accordingly not be read as evidence against an association, but as uninformative given the lack of reliable variance to explain.

Our findings suggest that anti-AI bias can extend to highly salient erotic stimuli, but only insofar as AI attribution is believed. Thus, the anti-AI bias is better understood as a belief-sensitive authenticity effect than as a uniform response to AI labels.

This result has relevant implications for research using synthetic stimuli. Attracted by the clear advantages in terms of controllability, scalability, and stimulus standardization, several researchers are promoting the use of AI-generated stimuli in psychological research (Albohn et al., 2022; Becker et al., 2024; Becker & Laycock, 2023; Chiu et al., 2026). However, their perceived origin must be controlled. Researchers using AI-generated erotic, affective, or social stimuli should therefore measure, manipulate, or report participants' beliefs about stimulus authenticity, rather than assuming that visual realism alone makes synthetic and photographic stimuli psychologically interchangeable.

Our results also corroborate the hypothesis that the 'authenticity aura' might indeed enhance the appeal of sexual content perceived to be 'genuine' over AI-generated content (Easterbrook-Smith, 2025). This might yield ambivalent effects. On the one hand, the general skepticism about content authenticity in the face of the rise of AI-generated content might dampen the incentives to spread non-consensual sexual material, as proposed by Viola and Voto (2023). On the other hand, this reduced affective reaction might translate into a 'moral discounting' and actually promote the spread of such content, based on the (false) assumption that explicitly fake non-consensual sexual deepfakes are not harmful (see Flynn et al., 2022).

A final consideration pertains to the possible impact of

disclosing deepfakes of sensual (though not necessarily sexual) pictures depicting human bodies in a wider societal context. Recall that some legislation, notably the recent EU AI Act (Art. 50), will soon mandate the explicit disclosure of the AI-generated origin of some media. An unexpected effect of such legislation when it comes, for instance, to advertising perfume or swimwear, might be that AI-generated models, while explicitly flagged as such, could lose some of their allure with respect to their human counterparts.

Interpreting the results of this study requires particular caution due to the nuance of its design. As expressed above, it is worth recalling that a substantial proportion of participants expressed doubts about the manipulation (in Study 1, 41.7% reported disbelieving the labels; in Study 2, 32% believed all images were real), meaning the experimental manipulation was not fully convincing for a meaningful portion of the sample. We contend, however, that this limitation extends to all similar studies, stressing the relevance of including explicit attention checks before drawing any conclusion.

Additionally, while the present findings are consistent with a crossover effect linking arousal and perceived authenticity, the design cannot establish the direction of this relationship. The observed congruency effects are equally consistent with authenticity impressions shaping subsequent arousal, or with arousal being retrospectively used to infer authenticity. Resolving this ambiguity would require designs capable of manipulating, rather than only measuring, one of these two processes. Moreover, future studies are needed to establish whether these effects extend to non-heterosexual samples.

To summarise, we investigated the anti-AI bias for erotic stimuli by conceptually replicating Marini et al. (2024). Participants viewed erotic (Studies 1 and 2) and non-erotic (Study 1) images, each preceded by a randomised cue indicating that the image was either a photograph or AI-generated. Across both studies, AI labelling produced consistent but small effects, driven primarily by male participants, alongside robust gender differences in responses to erotic stimuli that proved sensitive to relational context. Most importantly, affective responses depended on whether participants believed the labels: when realism judgements were congruent with the assigned label, images labelled as photographs were rated as more arousing than those labelled as AI-generated, whereas this pattern reversed when the labels were disbelieved. Individual differences in AI attitudes, pornography consumption, and sexual activity could not account for the labelling effect, as sensitivity to the AI label showed little reliable variability across participants. Together, these findings suggest that the anti-AI bias is best understood not as a uniform response to AI labels but as a belief-sensitive authenticity effect, one with practical implications for researchers using synthetic stimuli and for policies mandating the disclosure of AI-generated content.

References

- Ajder, H., Patrini, G., Cavalli, F., & Cullen, L. (2019). The state of deepfakes: Landscape, threats, and impact. *Amsterdam: Deeptrace*, 27.
- Albohn, D. N., Uddenberg, S., & Todorov, A. (2022). A data-driven, hyper-realistic method for visualizing individual mental representations of faces. *Frontiers in Psychology*, 13, 997498.
- Ansanı, A., Koehler, F., Giombini, L., Hämäläinen, M., Meng, C., Marini, M., & Saarikallio, S. (2025). AI performer bias: Listeners like music less when they think it was performed by an AI. *Empirical Studies of the Arts*, 43(2), 1137–1161.
- Becker, C., Conduit, R., Chouinard, P. A., & Laycock, R. (2024). Can deepfakes be used to study emotion perception? A comparison of dynamic face stimuli. *Behavior Research Methods*, 56(7), 7674–7690.
- Becker, C., & Laycock, R. (2023). Embracing deepfakes and AI-generated images in neuroscience research. *European Journal of Neuroscience*, 58(3), 2657–2661.
- Bellaiche, L., Shahi, R., Turpin, M. H., Ragnhildstveit, A., Sprockett, S., Barr, N., Christensen, A., & Seli, P. (2023). Humans versus AI: Whether and why we prefer human-created compared to AI-created artwork. *Cognitive Research: Principles and Implications*, 8(1), 42.
- Betker, J., Goh, G., Jing, L., Brooks, T., Wang, J., Li, L., Ouyang, L., Zhuang, J., Lee, J., Guo, Y., Manassra, W., Dhariwal, P., Chu, C., Jiao, Y., & Ramesh, A. (2023). *Improving image generation with better captions*. OpenAI. <https://cdn.openai.com/papers/dall-e-3.pdf>
- Bürkner, P.-C. (2017). Brms: An r package for bayesian multilevel models using stan. *Journal of Statistical Software*, 80, 1–28.
- Califano, G., & Spence, C. (2024). Assessing the visual appeal of real/AI-generated food images. *Food Quality and Preference*, 116, 105149.
- Carvalho, J., Czop, O., Rocha, M., Nobre, P., & Soares, S. (2018). Gender differences in the automatic attention to romantic vs sexually explicit stimuli. *The Journal of Sexual Medicine*, 15(8), 1083–1092.
- Cenerini, C., Keller, F., Pennazza, G., Santonico, M., & Vollero, L. (2025). Artistic turing test: The challenge of differentiating human and AI-generated art. *Computers in Human Behavior: Artificial Humans*, 100209.
- Chiarella, S. G., Torromino, G., Gagliardi, D. M., Rossi, D., Babiloni, F., & Cartocci, G. (2022). Investigating the negative bias towards artificial intelligence: Effects of prior assignment of AI-authorship on the aesthetic appreciation of abstract paintings. *Computers in Human Behavior*, 137, 107406.
- Chiu, H. T., Sou, H. I., Lam, Y. W., Ng, C. S. F., & Wong, S. W. (2026). Beyond traditional stimuli: Validating AI-generated images for eliciting negative emotions in affect research. *Plos One*, 21(2), e0342434.
- Chivers, M. L. (2017). The specificity of women's sexual response and its relationship with sexual orientations: A review and ten hypotheses. *Archives of Sexual Behavior*, 46(5), 1161–1179.
- Chivers, M. L., Rieger, G., Latty, E., & Bailey, J. M. (2004). A sex difference in the specificity of sexual arousal. *Psychological Science*, 15(11), 736–744.
- Ciesielski, B. G., Armstrong, T., Zald, D. H., & Olatunji, B. O. (2010). Emotion modulation of visual attention: Categorical and temporal characteristics. *PLoS One*, 5(11), e13860.
- Cloudy, J., Banks, J., & Bowman, N. D. (2022). Ai journalists and reduction of perceived hostile media bias: Replication and extension considering news organization cues. *Technology, Mind, and Behavior*, 3(3), 364.
- Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*. routledge.
- Cooke, D., Edwards, A., Barkoff, S., & Kelly, K. (2025). As good as a coin toss: Human detection of AI-generated content. *Communications of the ACM*, 68(10), 100–109.
- De Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a web browser. *Behavior Research Methods*, 47(1), 1–12.
- De Leeuw, J. R. (2024). DataPipe: Born-open data collection for online experiments. *Behavior Research Methods*, 56(3), 2499–2506.
- De Ruiter, A. (2021). The distinct wrong of deepfakes. *Philosophy & Technology*, 34(4), 1311–1332.
- Di Dio, C., Ardizzi, M., Schieppati, S. V., Massaro, D., Gilli, G., Gallese, V., & Marchetti, A. (2023). Art made by artificial intelligence: The effect of authorship on aesthetic judgments. *Psychology of Aesthetics, Creativity, and the Arts*.
- Döring, N., Le, T. D., Vowels, L. M., Vowels, M. J., & Marcantonio, T. L. (2024). The impact of artificial intelligence on human sexuality: A five-year literature review 2020–2024. *Current Sexual Health Reports*, 17(1), 4.
- Döring, N., & Mohseni, M. R. (2025). Anti-AI bias toward couple images and couple counseling: Findings from two experiments. *Archives of Sexual Behavior*, 1–14.
- Dunn, J. D., White, D., Sutherland, C. A., Miller, E. J., Steward, B. A., & Dawel, A. (2026). Too good to be true: Synthetic AI faces are more average than real faces and super-recognizers know it. *British Journal of Psychology*.
- Easterbrook-Smith, G. (2025). Pornographic aura, AI and the value of authenticity. *Porn Studies*, 1–13.
- Eiserbeck, A., Maier, M., Baum, J., & Abdel Rahman, R. (2023). Deepfake smiles matter less—the psychological and neural impact of presumed AI-generated faces. *Scientific Reports*, 13(1), 16111.
- Fisher, S. A. (2025). Something AI should tell you—the case

- for labelling synthetic content. *Journal of Applied Philosophy*, 42(1), 272–286.
- Fisher, S. A., Howard, J. W., & Kira, B. (2024). Moderating synthetic content: The challenge of generative AI. *Philosophy & Technology*, 37(4), 133.
- Flynn, A., Powell, A., Scott, A. J., & Cama, E. (2022). Deepfakes and digitally altered imagery abuse: A cross-country exploration of an emerging form of image-based sexual abuse. *The British Journal of Criminology*, 62(6), 1341–1358.
- Gangadharbatla, H. (2022). The role of AI attribution knowledge in the evaluation of artwork. *Empirical Studies of the Arts*, 40(2), 125–142.
- Grassini, S., & Koivisto, M. (2024). Understanding how personality traits, experiences, and attitudes shape negative bias toward AI-generated artworks. *Scientific Reports*, 14(1), 4113.
- Hatch, S. G., Esplin, C. R., Hatch, H. D., Halstead, A., Olsen, J., & Braithwaite, S. R. (2023). The consumption of pornography scale-general (COPS-g). *Sexual and Relationship Therapy*, 38(2), 194–218.
- Heiss, A. (2021). A guide to modeling proportions with bayesian beta and zero-inflated beta regression models. *Political Science*. See <https://www.andrewheiss.com/Blog/2021/11/08/Beta-Regression-Guide/> (accessed 8 February 2024).
- Hero, S. (2023). *State Of Deepfakes (2023): Realities, Threats, And Impact* | Security Hero — [securityhero.io](https://www.securityhero.io/state-of-deepfakes/). <https://www.securityhero.io/state-of-deepfakes/>.
- Holz, N., Larrouy-Maestri, P., & Poeppel, D. (2022). The variably intense vocalizations of affect and emotion (VI-VAE) corpus prompts new perspective on nonspeech perception. *Emotion*, 22(1), 213.
- Huang, M., Bridge, H., Kemp, M. J., & Parker, A. J. (2011). Human cortical activity evoked by the assignment of authenticity when viewing works of art. *Frontiers in Human Neuroscience*, 5, 134.
- Huberman, J. S., Maracle, A. C., & Chivers, M. L. (2015). Gender-specificity of women's and men's self-reported attention to sexual stimuli. *The Journal of Sex Research*, 52(9), 983–995.
- Itkes, O., & Kron, A. (2019). Affective and semantic representations of valence: A conceptual framework. *Emotion Review*, 11(4), 283–293.
- Köbis, N., & Mossink, L. D. (2021). Artificial intelligence versus maya angelou: Experimental evidence that people cannot differentiate AI-generated from human-written poetry. *Computers in Human Behavior*, 114, 106553.
- Kunaharan, S., Halpin, S., Sitharthan, T., Bosshard, S., & Walla, P. (2017). Conscious and non-conscious measures of emotion: Do they vary with frequency of pornography use? *Applied Sciences*, 7(5), 493.
- Kurdi, B., Lozano, S., & Banaji, M. R. (2017). Introducing the open affective standardized image set (OASIS). *Behavior Research Methods*, 49(2), 457–470.
- Laan, E., Everaerd, W., Bellen, G. van, & Hanewald, G. (1994). Women's sexual and emotional responses to male-and female-produced erotica. *Archives of Sexual Behavior*, 23(2), 153–169.
- Lang, P. J., Bradley, M. M., Cuthbert, B. N., et al. (1997). International affective picture system (IAPS): Technical manual and affective ratings. *NIMH Center for the Study of Emotion and Attention*, 1(39-58), 3.
- Lapointe, V. A., Dubé, S., Rukhlyadyev, S., Kessai, T., & Lafortune, D. (2025). The present and future of adult entertainment: A content analysis of AI-generated pornography websites. *Archives of Sexual Behavior*, 1–19.
- Liefvooghe, B., Oliveira, M., Leisten, L. M., Hoogers, E., Aarts, H., & Hortensius, R. (2023). Are natural faces merely labelled as artificial trusted less? *Collabra: Psychology*, 9(1), 73066.
- Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021). Performance: An r package for assessment, comparison and testing of statistical models. *Journal of Open Source Software*, 6(60).
- Lüdecke, D., Makowski, D., Ben-Shachar, M. S., Patil, I., Wiernik, B. M., Bacher, E., & Thériault, R. (2022). Easystats: Framework for easy statistical modeling, visualization, and reporting. *CRAN: Contributed Packages*.
- Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software*, 4(40), 1541.
- Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). Modelbased: An r package to make the most out of your statistical models through marginal means, marginal effects, and model predictions. *Journal of Open Source Software*, 10(109), 7969.
- Makowski, D., Te, A. S., Neves, A., Kirk, S., Liang, N. Z., Mavros, P., & Chen, S. A. (2025). Too beautiful to be fake: Attractive faces are less likely to be judged as artificially generated. *Acta Psychologica*, 252, 104670.
- Makowski, D., Wiernik, B. M., Patil, I., Lüdecke, D., & Ben-Shachar, M. S. (2022). *correlation: Methods for correlation analysis*. <https://CRAN.R-project.org/package=correlation>
- Marchewka, A., Żurawski, Ł., Jednoróg, K., & Grabowska, A. (2014). The nencki affective picture system (NAPS): Introduction to a novel, standardized, wide-range, high-quality, realistic picture database. *Behavior Research Methods*, 46(2), 596–610.
- Marini, M., Ansani, A., Demichelis, A., Mancini, G., Paglieri, F., & Viola, M. (2024). Real is the new sexy: The influence of perceived realness on self-reported arousal to sexual visual stimuli. *Cognition and Emotion*,

- 38(3), 348–360.
- McDonnell, R., & Breidt, M. (2010). Face reality: Investigating the uncanny valley for virtual faces. In *ACM SIGGRAPH ASIA 2010 sketches* (pp. 1–2).
- Miller, E. J., Steward, B. A., Witkower, Z., Sutherland, C. A., Krumhuber, E. G., & Dawel, A. (2023). AI hyperrealism: Why AI faces are perceived as more real than human ones. *Psychological Science*, 34(12), 1390–1403.
- Newman, G. E., & Bloom, P. (2012). Art and authenticity: The importance of originals in judgments of value. *Journal of Experimental Psychology: General*, 141(3).
- Nightingale, S. J., & Farid, H. (2022). AI-synthesized faces are indistinguishable from real faces and more trustworthy. *Proceedings of the National Academy of Sciences*, 119(8), e2120481119.
- Porter, B., & Machery, E. (2024). AI-generated poetry is indistinguishable from human-written poetry and is rated more favorably. *Scientific Reports*, 14(1), 26133.
- Prause, N., Steele, V. R., Staley, C., Sabatinelli, D., & Hajcak, G. (2015). Modulation of late positive potentials by sexual images in problem users and controls inconsistent with “porn addiction.” *Biological Psychology*, 109, 192–199.
- Ragot, M., Martin, N., & Cojean, S. (2020). Ai-generated vs. Human artworks. A perception bias towards artificial intelligence? *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–10.
- Rupp, H. A., & Wallen, K. (2007). Sex differences in viewing sexual stimuli: An eye-tracking study in men and women. *Hormones and Behavior*, 51(4), 524–533.
- Rupp, H. A., & Wallen, K. (2009). Sex-specific content preferences for visual sexual stimuli. *Archives of Sexual Behavior*, 38(3), 417–426.
- Safron, A., Sylva, D., Klimaj, V., Rosenthal, A., & Bailey, J. M. (2020). Neural responses to sexual stimuli in heterosexual and homosexual men and women: Men’s responses are more specific. *Archives of Sexual Behavior*, 49(2), 433–445.
- Schepman, A., & Rodway, P. (2020). Initial validation of the general attitudes towards artificial intelligence scale. *Computers in Human Behavior Reports*, 1, 100014.
- Schepman, A., & Rodway, P. (2023). The general attitudes towards artificial intelligence scale (GA AIS): Confirmatory validation and associations with personality, corporate distrust, and general trust. *International Journal of Human–Computer Interaction*, 39(13), 2724–2741.
- Shank, D. B., Stefanik, C., Stuhlsatz, C., Kacirek, K., & Belfi, A. M. (2023). AI composer bias: Listeners like music less when they think it was composed by an AI. *Journal of Experimental Psychology: Applied*, 29(3), 676.
- Stockner, M., Convertino, G., Cambedda, S., & Mazzoni, G. (2026). Are humans able to discriminate between real and deepfake faces? A systematic review and meta-analysis. *Computers in Human Behavior: Artificial Humans*, 100332.
- Tucciarelli, R., Vehar, N., Chandaria, S., & Tsakiris, M. (2022). On the realness of people who do not exist: The social processing of artificial faces. *Isience*, 25(12).
- Velásquez-Salamanca, D., Martín-Pascual, M. Á., & Andreu-Sánchez, C. (2025). Interpretation of AI-generated vs. Human-made images. *Journal of Imaging*, 11(7), 227.
- Viola, M., & Voto, C. (2023). Designed to abuse? Deepfakes and the non-consensual diffusion of intimate images. *Synthese*, 201(1), 30.
- Vuorre, M. (2019). How to analyze visual analog (slider) scale data. <https://Vuorre.Com/Posts/Analyze-Analog-Scale-Ratings-with-Zero-One-Inflated-Beta-Models/>.
- Walton, M. T., & Bhullar, N. (2018). Hypersexuality, higher rates of intercourse, masturbation, sexual fantasy, and early sexual interest relate to higher sexual excitation/arousal. *Archives of Sexual Behavior*, 47(8), 2177–2183.
- Wierzbna, M., Riegel, M., Pucz, A., Leśniewska, Z., Dragan, W. Ł., Gola, M., Jednoróg, K., & Marchewka, A. (2015). Erotic subset for the nencki affective picture system (NAPS ERO): Cross-sexual comparison study. *Frontiers in Psychology*, 6, 1336.
- Yang, W., Makita, K., Nakao, T., Kanayama, N., Machizawa, M. G., Sasaoka, T., Sugata, A., Kobayashi, R., Hiramoto, R., Yamawaki, S., et al. (2018). Affective auditory stimulus database: An expanded version of the international affective digitized sounds (IADS-e). *Behavior Research Methods*, 50(4), 1415–1429.
- Youn, G. (2006). Subjective sexual arousal in response to erotica: Effects of gender, guided fantasy, erotic stimulus, and duration of exposure. *Archives of Sexual Behavior*, 35(1), 87–97.
- Zakreski, E., Marečková, A., Vaníček, O., Hla, M., Klapilová, K., Lindová, J., & Pfau, J. G. (2026). Subjective responses of gynephilic men and women to real versus artificial female nudes. *Archives of Sexual Behavior*, 1–15.